NUMERICAL METHODS II

NUMERICAL SOLUTIONS OF PARTIAL DIFFERENTIAL EQUATIONS

PDE occur in many branches of applied mathematics, for example, in hydrodynamics, elasticity, quantum mechanics and electromagnetic theory. But only a small fraction of the PDE's occuring in science and engineering can be solved by formal analytical methods. In most cases, sufficiently approximate solution to applied PDE's can be got by simple and efficient numerical methods.

To solve numerically a PDE, we replace the derivatives appearing in the equation by finite-difference ratios. Then the D.E is replaced by a difference equation, which is solved by algebraic or arithmetic procedures.

CLASSIFICATION OF SECOND ORDER PARTIAL DIFFERENTIAL EQUATIONS

The general second order PDE in two independent variables is of the form

$$A\frac{\partial^2 u}{\partial x^2} + B\frac{\partial^2 u}{\partial x\,\partial y} + C\frac{\partial^2 u}{\partial y^2} + D\frac{\partial u}{\partial x} + E\frac{\partial u}{\partial y} + Fu = G$$

which can be written as

$$A u_{xx} + B u_{xy} + C u_{yy} + D u_x + E u_y + F u = G \quad \text{——— (1)}$$

where A, B, C, D, E, F, G are functions of x and y only.

The above eqⁿ is said to be

(i) Elliptic at a point (x, y) in the plane if

$$B^2 - 4AC < 0$$

(ii) Parabolic if $B^2-4AC=0$

and (iii) Hyperbolic if $B^2-4AC>0$

Ex:- (i) $f_{xx}+2f_{xy}+f_{yy}=0$

Here $A=1,\ B=2,\ C=1$

$B^2-4AC = 4-4(1)(1)=0$

The equation is Parabolic

(ii) $f_{xy}-f_x=0$

Here $A=0,\ B=1,\ C=0$

$B^2-4AC=1>0 \Rightarrow$ Equation is Hyperbolic

(iii) $y\,u_{xx}+u_{yy}=0$

Here $A=y,\ B=0,\ C=1$

$B^2-4AC = 0-4(y)(1) = -4y$

The given equation is Elliptic for $y>0$

Parabolic for $y=0$

Hyperbolic for $y<0$

(iv) $x^2 f_{xx}+(1-y^2)f_{yy}=0$, $-\infty<x<\infty$, $-1<y<1$

Here $A=x^2,\ B=0,\ C=1-y^2$

$B^2-4AC = 0-4x^2(1-y^2) = 4x^2(y^2-1)$

for all values of x between $-\infty$ and ∞, x^2 is +ve

As y lies b/w -1 & 1, y^2 is <1

Hence B^2-4AC is -ve in the given region

so the equation is Elliptic

FINITE DIFFERENCE APPROXIMATIONS TO PARTIAL DERIVATIVES

We know that the derivatives of a function of a single variable is defined as

$$\frac{dy}{dx} = \lim_{h \to 0} \frac{y(x+h) - y(x)}{h}$$

In a digital computer, we cannot take limit. So we say that

$$\frac{dy}{dx} = \frac{y(x+h) - y(x)}{h} \quad \text{where } h \text{ is small}$$

Thus the derivative is replaced by a difference quotient. We make the same approach to partial differential equations.

By Taylor's formula

$$u(x+h, y) = u(x,y) + h\, u_x(x,y) + \frac{h^2}{2!} u_{xx}(x,y) + \frac{h^3}{3!} u_{xxx}(x,y) - \cdots \quad \text{——— (1)}$$

$$u(x-h, y) = u(x,y) - h\, u_x(x,y) + \frac{h^2}{2!} u_{xx}(x,y) - \frac{h^3}{3!} u_{xxx}(x,y) - \cdots \quad \text{——— (2)}$$

(1) − (2), Neglecting terms in $h^3, h^4 \ldots$ and solve for

$$u_x(x,y) = \frac{1}{2h} \left[u(x+h, y) - u(x-h, y) \right]$$

Simillarly

$$u(x, y+k) = u(x,y) + k\,u_y(x,y) + \frac{k^2}{2!}u_{yy}(x,y) + \cdots \quad \text{——(3)}$$

$$u(x, y-k) = u(x,y) - k\,u_y(x,y) + \frac{k^2}{2!}u_{yy}(x,y) + \cdots \quad \text{——(4)}$$

(3)-(4), Neglecting terms in $k^3, k^4 \ldots$ and Solving for u_y we obtain

$$u_y(x,y) = \frac{1}{2k}\left[u(x, y+k) - u(x, y-k)\right]$$

For second derivatives

(1) + (2) $\Rightarrow$

$$u_{xx}(x,y) = \frac{1}{h^2}\left[u(x+h, y) - 2u(x,y) + u(x-h,y)\right]$$

Similarly

$$u_{yy}(x,y) = \frac{1}{k^2}\left[u(x, y+k) - 2u(x,y) + u(x, y-k)\right]$$

FINITE DIFFERENCE ANALOGUES FOR PARTIAL DERIVATIVES

$$u_x = \frac{u_{i+1,j} - u_{i,j}}{h} \quad \text{(forward difference)}$$

$$= \frac{u_{i,j} - u_{i-1,j}}{h} \quad \text{(backward difference)}$$

$$u_y = \frac{u_{i,j+1} - u_{i,j}}{h} \quad \text{(forward difference)}$$

$$= \frac{u_{i,j} - u_{i,j-1}}{h} \quad \text{(backward difference)}$$

SOLUTION OF LAPLACE EQUATION IN TWO DIMENSION

Laplace equation in two dimension is given by

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad \text{or} \quad \nabla^2 u = 0$$

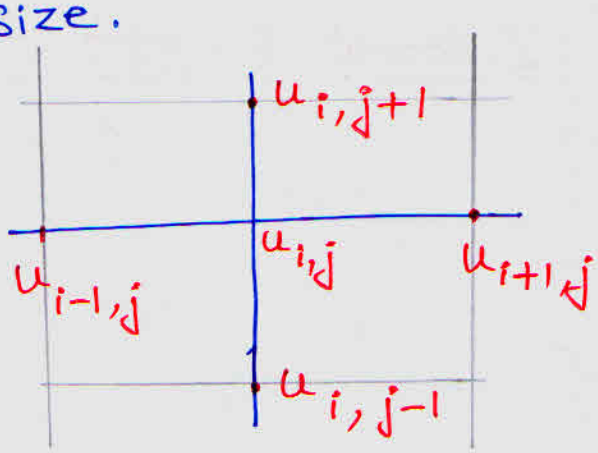
The difference equation for Laplace equation is ($h = k$)

$$\frac{1}{h^2}\left[u(x+h, y) + u(x, y+h) + u(x-h, y) + u(x, y-h) - 4u(x, y)\right] = 0$$

(or)

$$u(x+h, y) + u(x, y+h) + u(x-h, y) + u(x, y-h) - 4u(x, y) = 0 \quad \text{---} (5)$$

h is called the mesh size.



We adopt the notation

$$u(x, y) = u_{i,j}$$

$$u(x+h, y) = u_{i+1,j}$$

$$u(x-h, y) = u_{i-1,j}$$

$$u(x, y+h) = u_{i,j+1}$$

$$u(x, y-h) = u_{i,j-1}$$

Eq^n (5) becomes

$$u_{i+1,j} + u_{i,j+1} + u_{i-1,j} + u_{i,j-1} - 4u_{i,j} = 0$$

$$\Rightarrow u_{i,j} = \frac{1}{4}\left[u_{i+1,j} + u_{i,j+1} + u_{i-1,j} + u_{i,j+1}\right] \quad \text{---------- (1)}$$

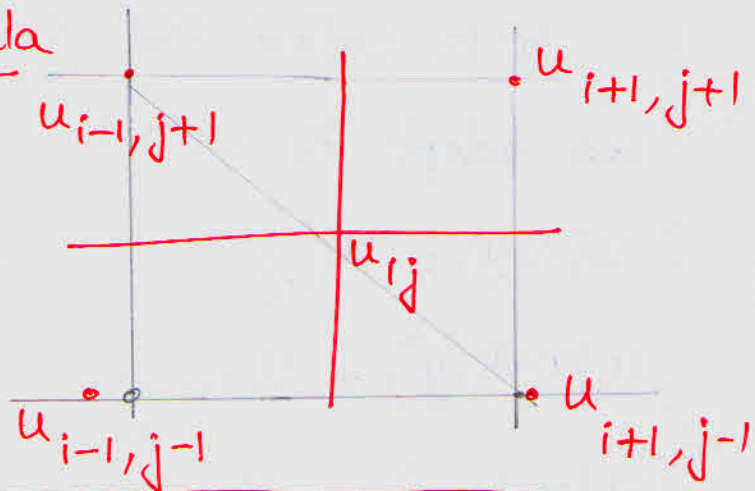
This shows that the value of $u_{i,j}$ at any interior mesh point is the average of its value at four neighbouring points to the left, right, above and below.

Eq 1

$$\boxed{u_{i,j} = \frac{1}{4}\left[u_{i+1,j} + u_{i,j+1} + u_{i-1,j} + u_{i,j-1}\right]}$$

is called the Standard 5 point formula

Diagonal 5-point formula



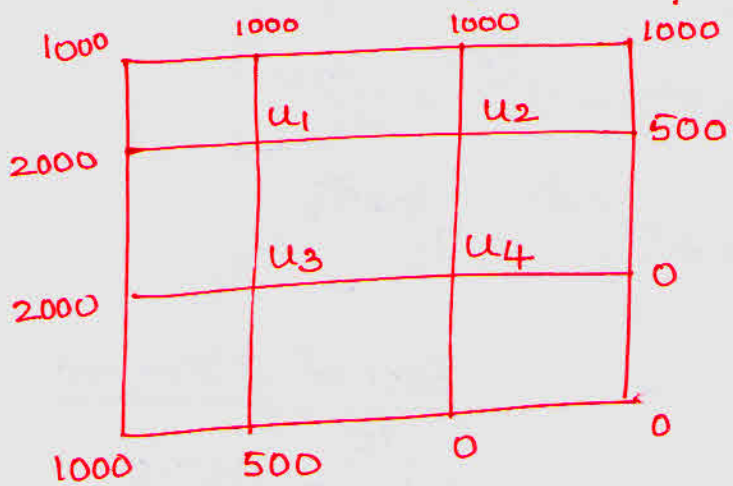
$$\boxed{u_{i,j} = \frac{1}{4}\left[u_{i-1,j+1} + u_{i+1,j-1} + u_{i+1,j+1} + u_{i-1,j-1}\right]}$$

Gauss-Seidal method which when applied to PDE is also referred to as Liebmann's method.

Note :- Diagonal 5-point formula is valid since Laplace eqⁿ remains invariant when the coordinate axes are rotated through 45°. It can be shown that the error in the diagonal formula is four times than in the standard formula. Hence in all computations, we use the standard formula instead of the diagonal formula whenever possible.

PROBLEMS

1. Given the values of $u(x,y)$ on the boundary of the square region. Evaluate the function $u(x,y)$ satisfying the Laplace equation $\nabla^2 u = 0$ at the pivotal points of this figure



Solution :- To get initial values of u_1, u_2, u_3 and u_4

we assume $u_4 = 0$

$$u_1 = \frac{1}{4}\left[1000 + u_4 + 1000 + 2000\right] = 1000 \quad \text{[Diagonal 5 point formula]}$$

$$u_2 = \frac{1}{4}\left[1000 + u_4 + u_1 + 500\right] = \frac{1}{4}\left[1000 + 0 + 1000 + 500\right]$$

$$= 625 \quad \text{[standard 5 point]}$$

$$u_3 = \frac{1}{4}\left[u_1 + 500 + u_4 + 2000\right] = \frac{1}{4}\left[1000 + 500 + 0 + 2000\right]$$

$$= 875 \quad \text{(std)}$$

$$u_4 = \frac{1}{4}\left[u_3 + 0 + 0 + u_2\right] = \frac{1}{4}\left[875 + 625\right] = 375 \quad \text{(std)}$$

$$u_1^{(n+1)} = \frac{1}{4}\left[2000 + 1000 + u_3^{(n)} + u_2^{(n)}\right]$$

$$u_2^{(n+1)} = \frac{1}{4}\left[1000 + 500 + u_1^{(n+1)} + u_4^{(n)}\right]$$

$$u_3^{(n+1)} = \frac{1}{4}\left[2000 + 500 + u_4^{(n)} + u_1^{(n+1)}\right]$$

$$u_4^{(n+1)} = \frac{1}{4}\left[0 + 0 + u_3^{(n+1)} + u_2^{(n+1)}\right]$$

First Iteration

$u_1^{(1)} = 1125$

$u_2^{(1)} = 750$

$u_3^{(1)} = 1000$

$u_4^{(1)} = 437.5$

Second Iteration

$u_1^{(2)} = 1187.5$

$u_2^{(2)} = 781.25$

$u_3^{(2)} = 1031.25$

$u_4^{(2)} = 453.125$

Third Iteration

$u_1^{(3)} = 1203.125$

$u_2^{(3)} = 789.0625$

$u_3^{(3)} = 1039.0625$

$u_4^{(3)} = 457.03125$

Fourth Iteration

$u_1^{(4)} = 1207.03$

$u_2^{(4)} = 791.01$

$u_3^{(4)} = 1041.01$

$u_4^{(4)} = 458.00$

Fifth Iteration

$u_1^{(5)} = 1208.00$

$u_2^{(5)} = 791.50$

$u_3^{(5)} = 1041.50$

$u_4^{(5)} = 458.25$

Sixth Iteration

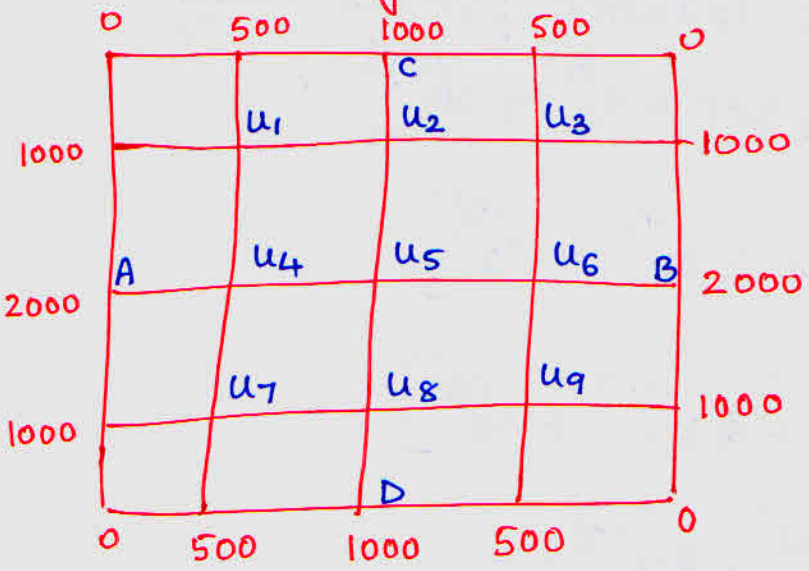
$u_1^{(6)} = 1208.25$

$u_2^{(6)} = 791.62$

$u_3^{(6)} = 1041.62$

$u_4^{(6)} = 458.31$

(2) Solve the Elliptic equation $u_{xx} + u_{yy} = 0$ for the Square mesh with boundary values as shown



Solution :- Let $u_1, u_2, \ldots u_9$ be the values of u at the interior mesh points. Since the boundary values of u are symmetrical about AB

$\therefore u_1 = u_7$, $u_2 = u_8$, $u_3 = u_9$

Also the values of u being symmetrical about CD

$u_3 = u_1$, $u_6 = u_4$, $u_9 = u_7$

Thus it is sufficient to find the values u_1, u_2, u_4 and u_5.

using standard Five point formula

$$u_5 = \frac{1}{4}\left[2000 + 2000 + 1000 + 1000\right] = 1500$$

using Diagonal five point formula

$$u_1 = \frac{1}{4}\left[0 + 1500 + 1000 + 2000\right] = 1125$$

using standard five point formula

$$u_2 = \frac{1}{4}\left[1125 + 1125 + 1000 + 1500\right] = 1187.5$$

$$u_4 = \frac{1}{4}\left[1125 + 1125 + 2000 + 1500\right] = 1437.5$$

We carry out the iteration process using

$$u_1^{(n+1)} = \frac{1}{4}\left[1000 + 500 + u_4^{(n)} + u_2^{(n)}\right]$$

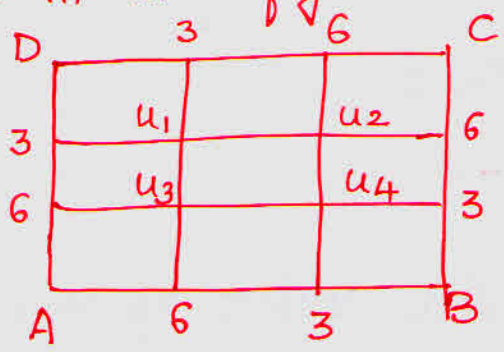
$$u_2^{(n+1)} = \frac{1}{4}\left[1000 + 2u_1^{(n)} + u_5^{(n)}\right]$$

$$u_4^{(n+1)} = \frac{1}{4}\left[2000 + 2u_1^{(n)} + u_5^{(n)}\right]$$

$$u_5^{(n+1)} = \frac{1}{4}\left[2u_4^{(n)} + 2u_2^{(n)}\right]$$

Iteration	u_1	u_2	u_4	u_5
1	1031.25	1140.62	1390.62	1265.62
2	1007.8	1070.31	1320.31	1195.31
3	972.65	1035.15	1285.15	1160.15
4	955.07	1017.57	1267.57	1142.57
5	946.28	1008.78	1258.78	1133.78
6	941.89	1004.39	1254.39	1129.39
7	939.69	1002.19	1252.19	1127.19
8	938.59	1001.09	1251.09	1126.09
9	938.04	1000.54	1250.54	1125.54

3 Solve $u_{xx} + u_{yy} = 0$ numerically for the following mesh with uniform spacing and with boundary conditions as shown below in the figure



Solution :- We note that the PDE and the boundary conditions are symmetric about the diagonals AC and BD

Hence $u_1 = u_4$ and $u_2 = u_3$

∴ We need to solve for two unknowns u_1 & u_2

$$u_1 = \frac{1}{4}\left[3 + 3 + u_3 + u_2\right] \Rightarrow 4u_1 = 6 + 2u_2$$

$$2u_1 - u_2 = 3 \quad \text{—(1)}$$

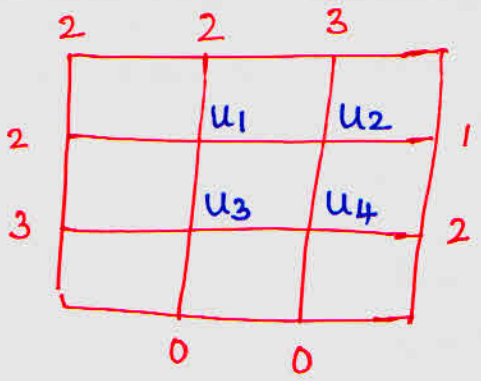
$$u_2 = \frac{1}{4}\left[6 + 6 + u_1 + u_4\right] \Rightarrow 4u_2 = 12 + 2u_1$$

$$u_1 - 2u_2 = -12 \quad \text{—(2)}$$

Solving (1) & (2), we get

$u_1 = 4$ and $u_2 = 5$

(4) Solve $u_{xx} + u_{yy} = 0$ numerically, using five point formula and Liebmann method, for the following mesh with uniform spacing



Solution :- To get the initial values of u_1, u_2, u_3, u_4

we assume $u_4 = 0$

$$u_1 = \frac{1}{4}[2+3+3+0] = 2$$

$$u_2 = \frac{1}{4}[2+1+3+0] = 1.5$$

$$u_3 = \frac{1}{4}[2+3+0+0] = 1.25$$

$$u_4 = \frac{1}{4}[1.25+1.5+2+0] = 1.1875$$

We carry out iteration process using

$$u_1 = \frac{1}{4}[2+2+u_3+u_2] \Rightarrow u_1 = 0.25[4+u_3+u_2]$$

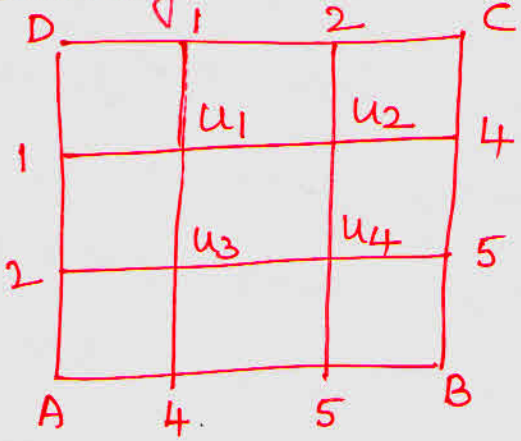
$$u_2 = \frac{1}{4}[3+u_1+u_4+1] \Rightarrow u_2 = 0.25[4+u_1+u_4]$$

$$u_3 = \frac{1}{4}[3+0+u_1+u_4] \Rightarrow u_3 = 0.25[3+u_1+u_4]$$

$$u_4 = \frac{1}{4}[2+0+u_2+u_3] \Rightarrow u_4 = 0.25[2+u_2+u_3]$$

Iteration	u_1	u_2	u_3	u_4
1	1.6875	1.7187	1.4687	1.2968
2	1.7968	1.7734	1.5234	1.3242
3	1.8242	1.7871	1.5371	1.3310
4	1.8310	1.7905	1.5405	1.3327
5	1.8327	1.7913	1.5413	1.3331

5 Using the Liebmann method, solve the equation $u_{xx} + u_{yy} = 0$ for the following square mesh with boundary values as shown in figure



Solution:- The boundary values are symmetric with respect to line AC. Hence $u_2 = u_3$

∴ We have three unknowns u_1, u_2 and u_4

$$u_1 = \frac{1}{4}[1+1+u_2+u_3] \Rightarrow u_1 = 0.25[2+2u_2]$$

$$u_1 = 0.5[1+u_2]$$

$$u_2 = \frac{1}{4}[2+4+u_1+u_4] \Rightarrow u_2 = 0.25[6+u_1+u_4]$$

$$u_4 = \frac{1}{4}[u_2+u_3+5+5] \Rightarrow u_4 = 0.5[5+u_2]$$

Taking initial values $u_2 = 0$ and $u_4 = 0$

Iteration	u_1	u_2	u_3
1	0.5	1.625	3.3125
2	1.3125	2.6562	3.8281
3	1.8281	2.9140	3.9570
4	1.9570	2.9785	3.9892
5	1.9892	2.9946	3.9973
6	1.9973	2.9986	3.9993

6 Solve $\nabla^2 u = 0$ in the square region bounded by the coordinate axes and the lines $x = 4$ and $y = 4$ with the boundary conditions given by the expressions

$$u(0, y) = 0 \quad \text{for } 0 \le y \le 4$$

$$u(4, y) = 12 + y \quad \text{for } 0 \le y \le 4$$

$$u(x, 0) = 3x \quad \text{for } 0 \le x \le 4$$

$$u(x, 4) = x^2 \quad \text{for } 0 \le x \le 4$$

Also employ Liebmann's iteration process to compute the second iterative values of $u(x, y)$.

Solution :- We shall divide the square region into 16 squares of side one unit ($h = 1$)

We shall derive the values of $u(x, y)$ on the boundary from given expression

(i) $u(0,y) = 0 \Rightarrow u(0,1) = 0,\ u(0,2) = u(0,3) = u(0,4) = 0$

(ii) $u(4,y) = 12 + y \Rightarrow u(4,1) = 13,\ u(4,2) = 14,\ u(4,3) = 15$

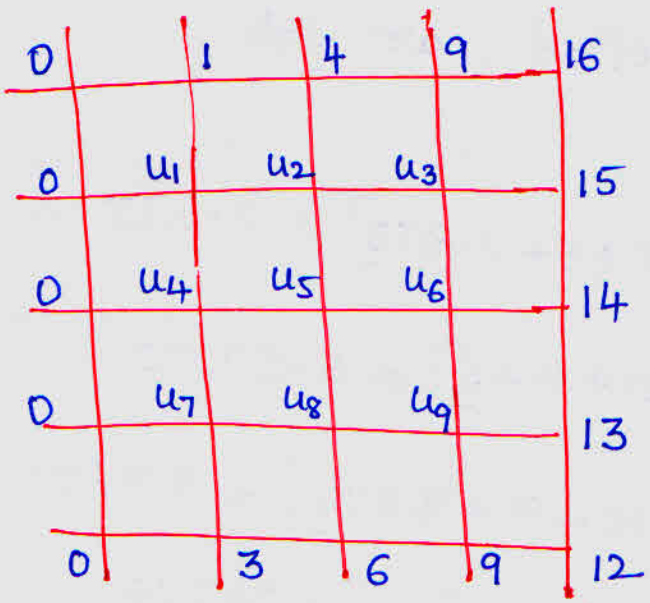
$u(4,4) = 16$

(iii) $u(x,0) = 3x \Rightarrow u(0,0) = 0,\ u(1,0) = 3,\ u(2,0) = 6$

$u(3,0) = 9 \quad u(4,0) = 12$

(iv) $u(x,4) = x^2 \Rightarrow u(1,4) = 1,\ u(2,4) = 4,\ u(3,4) = 9$

$u(4,4) = 16$



$u_5 = \frac{1}{4}[4 + 0 + 14 + 6] = 6$ (SFPF)

$u_1 = \frac{1}{4}[0 + 4 + 0 + 6] = 2.5$

$u_3 = \frac{1}{4}[4 + 16 + 6 + 14] = 10$

$u_7 = \frac{1}{4}[0 + 6 + 0 + 6] = 3$

$u_9 = \frac{1}{4}[6 + 14 + 6 + 12] = 9.5$

(DFPF)

Using S.F.P.F

$$u_2 = \frac{1}{4}[4+6+2.5+10] = 5.625$$

$$u_4 = \frac{1}{4}[0+2.5+3+6] = 2.875$$

$$u_6 = \frac{1}{4}[6+9.5+14+10] = 9.875$$

$$u_8 = \frac{1}{4}[3+9.5+6+6] = 6.125$$

Using these values as initial approximation to start the Liebmann's method, we get

First iteration

$$u_1 = \frac{1}{4}[0+5.625+1+2.875] = 2.375$$

$$u_2 = \frac{1}{4}[2.375+10+4+6] = 5.59375$$

$$u_3 = \frac{1}{4}[5.59375+15+9+9.875] = 9.8671$$

$$u_4 = \frac{1}{4}[0+6+2.375+3] = 2.84375$$

$$u_5 = \frac{1}{4}[2.84375+9.875+5.59375+6.125] = 6.10937$$

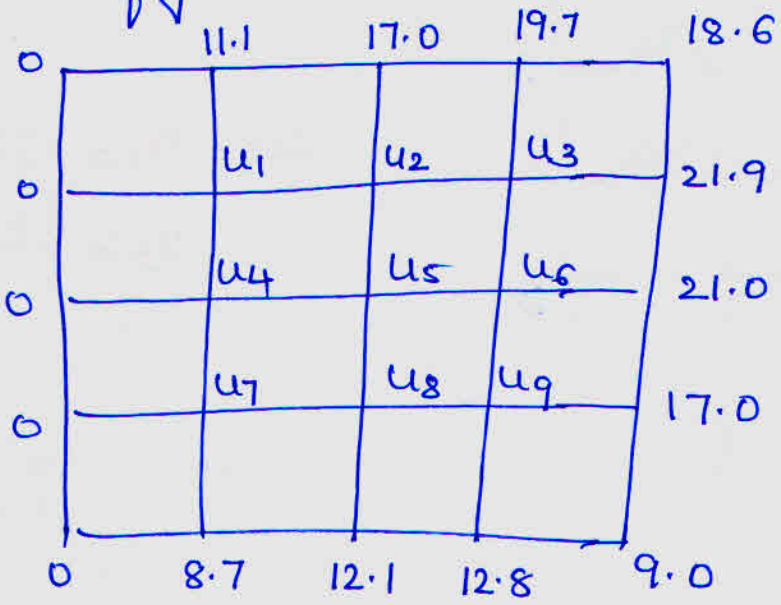
$$u_6 = \frac{1}{4}[6.10937+14+9.86718+9.5] = 9.8691$$

$$u_7 = \frac{1}{4}[0+6.125+2.8437+3] = 2.9921$$

$$u_8 = \frac{1}{4}[2.9921+9.5+6.1093+6] = 6.1503$$

$$u_9 = \frac{1}{4}[6.1503+13+9.8691+9] = 9.5048$$

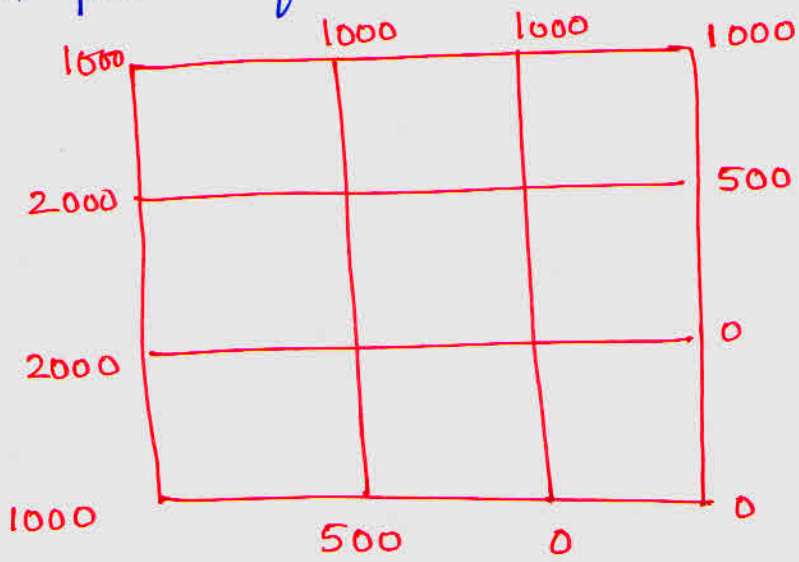
7. Determine by iteration method the values at the interior lattice points of a square region of the harmonic function u whose boundary values are given as shown in the figure.



Ans:- $u_1 = 7.9$, $u_2 = 13.7$, $u_3 = 17.9$, $u_4 = 6.6$

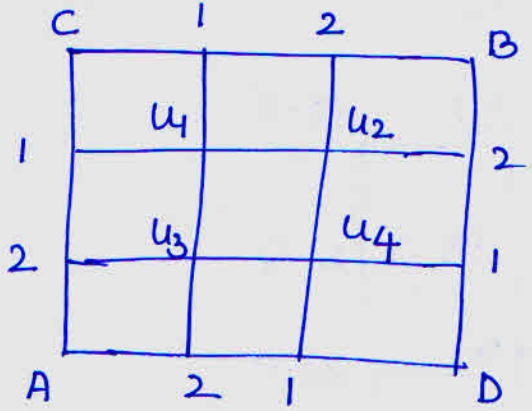
$u_5 = 11.9$, $u_6 = 16.3$, $u_7 = 6.6$, $u_8 = 11.2$, $u_9 = 14.3$

(8) Given the values of $u(x,y)$ on the boundary of the square given in the figure, evaluate the function $u(x,y)$ satisfying Laplaces equation $\nabla^2 u = 0$ at the pivotal points of this figure



Ans:- $u_1 = 1202$, $u_2 = 792$, $u_3 = 1042$, $u_4 = 459$

9. Solve the elliptic equation $u_{xx} + u_{yy} = 0$ for the following Square mesh with boundary values as shown, using Liebmann method.



Ans:- $u_1 = 1.333 = u_4$

$u_2 = 1.667 = u_3$